# EXAM SOLUTIONS

1. a) i) [ 3 ]

Many students got confused in this question and instead of digital vs. analog, they identified AM vs. FM as the two major modulation schemes, although the two have the same performance measure, SNR.

Modulation is the modification of the source signals at the transmitter into a form that is suitable to be transmitted over the channel.

We have studied analog and digital modulation schemes. Analog modulation maps a continuous signal to a parameter of the carrier waveform continuously; whereas in digital communication, we first discretize the message signal, and map the corresponding values to discrete values of a paramater of the carrier waveform.

We use signal-to-noise ratio (SNR) to measure the performance of analog modulation, and error probability for digital communications.

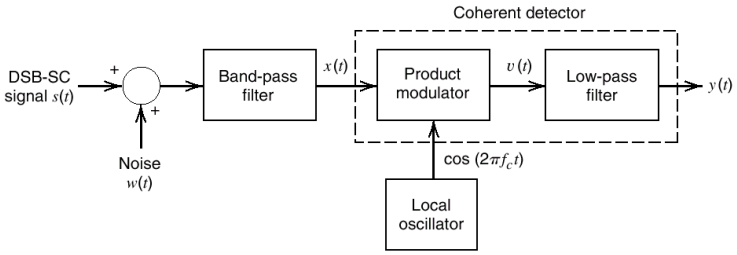
ii) [ 3 ]

The complex envelope of $x(t)$ is given by $\tilde{x}(t) = x_I(t) + jx_Q(t)$, and we have

$$x(t) = \text{Re}\{\tilde{x}(t)e^{j2\pi f_c t}\}.$$

iii) [ 4 ]

I was glad to see that almost all students managed to answer this question correctly.




Bandpass filter at the front end removes the out-of-band noise. We then multiply the filtered received signal with $2\cos(2\pi f_c t)$ to obtain a baseband signal corresponding to the in-phase component. Low-pass filter removes the high frequency components that are located around $\pm 4f_c$.

iv) [ 2 ]

SSB uses half the bandwidth of DSB-SC while achieving the same SNR performance.

b) i) True. [ 1 ]

ii) True. [ 1 ]

iii) False. $S_X(f) = S_X(-f), \ \forall\, f.$ [ 1 ]

iv) False. $R_X(0) = \int_{-\infty}^{+\infty} S_X(f)df.$ [ 1 ]

v) True. $E[Y(t)] = H(0)E[X(t)] = 0$, where $H(\cdot)$ is the Fourier transform corresponding to Hilbert transform, which is given by $-j\text{sgn}(f)$. [ 2 ]

Students did reasonably well in general. The last part was the hardest one. Some students could not remember the Hilbert transform, while some could not get its Fourier transform correctly, although it was provided in the Table on page 1.

c) i) [ 3 ]

We have $Y \sim \mathcal{N}(-4,4)$. We can write

$$\Pr(Y > 3) = \Pr((Y+4)/2 > 3.5) = \Pr(\bar{Y} > 3.5),$$

where $\bar{Y}$ is a standard normal distribution. Then we have $\Pr(Y > 3) = Q(3.5)$.

Some students got confused in converting the integral expression to Q function. There were also surprisingly many mistakes in the last part. Many students did not recall that the noise is still Gaussian (since it is a linear combination of Gaussian), and many had problem in evaluating its variance.

ii) [ 4 ]

We have $Y \sim \mathcal{N}(0,4)$. We can write

$$\Pr(3 \le Y \le 10) = \Pr(1.5 \le Y/2 \le 5) = \Pr(1.5 \le \bar{Y} \le 5),$$

where $\bar{Y}$ is a standard normal distribution. We have $\Pr(3 \le Y \le 10) = Q(1.5) - Q(5)$.

iii) [ 5 ]

We have $Y \sim \mathcal{N}(1,25)$. We can write

$$\Pr(Y \le 4) = \Pr((Y-1)/5 \le 4) = \Pr(\bar{Y} \le 0.6),$$

where $\bar{Y}$ is a standard normal distribution. Then we have $\Pr(Y \le 4) = \Pr(\bar{Y} \ge -0.6) = Q(-0.6)$.

d) i) [ 2 ]

$P_S = A^2 + \frac{B^2}{2}$.

ii) [ 2 ]

Quantization noise is assumed to have a uniform distribution with pdf $f_Q(q) = \frac{1}{\Delta}$ for $-\frac{\Delta}{2} < f \le \frac{\Delta}{2}$, where $\Delta$ is the length of the quantization interval. It follows that the quantization noise variance is given by $P_N = \Delta^2/12$.

Almost all students evaluated the power correctly, and wrote the pdf and variance of the quantization noise. There were some issues in finding the dynamic range of the signal; many wrote (2A +2B) instead of 2B. The average performance was good in this question.

The signal range is $[A-B, A+B]$. Hence, we have

$$\Delta = \frac{2B}{2^n} = \frac{B}{2^{n-1}}.$$

iii) [ 3 ]

$$\begin{aligned} SNR &= \frac{P_S}{P_N} \\ &= \frac{A^2 + \frac{B^2}{2}}{\frac{B^2}{12 \cdot 2^{2n-2}}} \\ &= 3\left(\frac{A^2}{B^2} + \frac{1}{2}\right) 2^{2n}. \end{aligned}$$

Then

$$SNR_{dB} = 6.02n + 4.77 + 10 \cdot \log_{10}\left(\frac{A^2}{B^2} + \frac{1}{2}\right).$$

iv) [ 3 ]

We need $n \ge 15$ to have $SNR \ge 100\ dB$.

2. a) i) [ 4 ]

We obtain the following codewords for each symbol pair

| Symbol | Probability | Codeword |
|---|---|---|
| cz | 7/32 | 10 |
| dz | 6/32 | 11 |
| dy | 5/32 | 001 |
| dx | 4/32 | 010 |
| bt | 4/32 | 011 |
| by | 3/32 | 0000 |
| cx | 2/32 | 00010 |
| ax | 1/32 | 00011 |

The average codeword length is $92/32 = 2.875$.

Most students did well in deriving the Huffman code. There were some confusion in obtaining the codewords, but the general mechanism was run successfully by many. There were less correct answers for part 2. Some student simply added the average lengths of the two codebooks, without taking the probabilities into account.

ii) [ 5 ]

If we consider each source separately, we obtain the following codes:

| | Source $S_1$ | |
|---|---|---|
| Symbol | Probability | Codeword |
| d | 15/32 | 1 |
| c | 9/32 | 00 |
| b | 7/32 | 010 |
| a | 1/32 | 011 |

The average codeword length per source symbol for source $S_1$ is $57/32 = 1.7813$ bits.

| | Source $S_2$ | |
|---|---|---|
| Symbol | Probability | Codeword |
| z | 13/32 | 1 |
| y | 8/32 | 01 |
| x | 7/32 | 000 |
| t | 4/32 | 001 |

The average codeword length per source symbol for source $S_2$ is $62/32 = 1.9375$ bits.

We find the average code length for the proposed code by averaging the sum code length for each symbol pair. We have

$$\sum_{s_1 \in \mathscr{S}_1, s_2 \in \mathscr{S}_2} p(s_1, s_2)(\text{length}(s_1) + \text{length}(s_2)),$$

where $\text{length}(s_i)$ is the length of the corresponding codeword for each symbol obtained from the two separate codes. We find the average length per symbol pair as $119/32 = 3.719$.

iii) [ 4 ]

Joint encoding achieves a lower average code length. The answer would still be the same if the sources were independent; this is the idea behind Shannon's source coding theorem.

Many students missed this point, although we have definitely gone through Shannon's source coding theorem, and the idea of typical sequences.

iv) [ 5 ]

Entropy calculation was fine overall, but there was some confusion about its consequence on practical codes.

$H(S_1,S_2) = 2.8273$. This is smaller than the average length of both codes. The average code length reduces as we code over more and more source symbol pairs.

No, entropy is a lower bound on the minimum average code length for any prefix code.

b) For a real WSS process, we have

$$E\left[\Big(X(t+\tau)-X(t)\Big)^2\right] = E[X^2(t+\tau)-2X(t+\tau)X(t)+X^2(t)]$$
$$= 2R_X(0)-2R_X(\tau)$$
$$= 2[R_X(0)-R_X(\tau)].$$

i) [ 4 ]

For $\tau = 3$, we have

$$2[R_X(0)-R_X(\tau)] = 2[A-Ae^{-3\alpha}+1].$$

I was expecting to see many more correct answers to this question. Even those who answered it correctly did not see that the first two parts are essentially two special cases of the same problem.

ii) [ 3 ]

For $\tau = -3$, we get the same result as above since $R_X(\tau)$ is an even function.

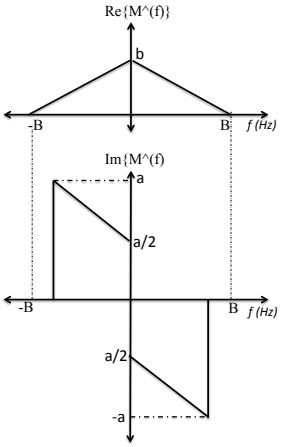
iii) [ 5 ]

$$E\left[\Big(X(t+\tau)+X(t)-X(t-\tau)\Big)^2\right] = E[X^2(t+\tau)+X^2(t)+X^2(t-\tau)+2X(t+\tau)X(t)$$
$$-2X(t+\tau)X(t-\tau)-2X(t)X(t-\tau)]$$
$$= 3R_X(0)-2R_X(2\tau)$$
$$= 3A-2(Ae^{-6\alpha}-2)$$

3. a) i) [ 6 ]

We have $\hat{M}(f) = -j \cdot sgn(f) \cdot M(f)$

While a good number of students wrote down the Fourier transform correctly, they had problems in plotting it.
In the second part, some students automatically plotted the SSB-USB signal, since that was the one we did in the class.



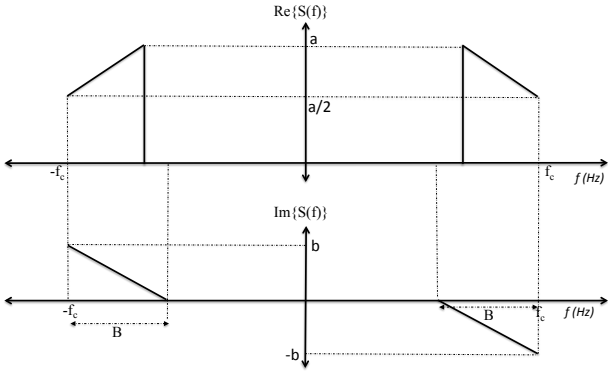

ii) [ 8 ]

The students may remember from class that the USB-LSB transmits only the lower halves of the spectrum of the signal, centered around carrier frequency $f_c$.

They should also be able to derive this directly from the basic principles as follows:

$$S(f) = \frac{1}{2}\left[M(f-f_c) + M(f+f_c)\right] + \frac{1}{2j}\left[\hat{M}(f-f_c) - \hat{M}(f+f_c)\right]$$

By simply shifting the spectrum of $M(f)$ and $\hat{M}(f)$ and applying the appropriate sign and conjugate adjustments they get the spectrum as below.




iii) [ 2 ]

I was surprised to see that some students answered this as 2B despite answering question 1/a/iv) correctly.

We need a filter with bandwidth $B$, which passes only frequencies between $[-f_c, -f_c + B]$ and $[f_c - B, f_c]$.

Part iv was also answered by a surprisingly small number of students. I assume this was partially due to timing; many did not even have the time to try, but many others delved into much more complicated derivations.

iv) [ 4 ]

Multiplication with $\cos(2\pi f_c t + 3\pi/2) = \sin(2\pi f_c t)$ will decode the quadrature component of the modulated signal. Therefore, after low pass filtering, we will obtain $\hat{m}(t)$.

Since Hilbert transform is one to one, we can recover the message signal, by taking the Hilbert transform of $\hat{m}(t)$, we get $-m(t)$.

b) i) [ 2 ]

Let $X$ denote the input bit, and $Y$ denote the decoded bit.

$$P(X=0|Y=1) = \frac{P(X=0,Y=1)}{P(Y=1)} = \frac{0.5(1-p_0)}{0.5(1-p_0+p_1)} = \frac{1-p_0}{1-p_0+p_1}.$$

ii) [ 3 ]

Probability of error is minimized by mapping the more likely outcome to the more reliable bit. We transmit a 0 for tails, and a 1 for heads. The probability of error is then given by $0.7(1-p_0)+0.3(1-p_1) = 1-0.7p_0-0.3p_1$.

This might be the question from which the students received the smallest average mark. Again, I assume this was partially due to lack of time, but also because it is quite different from standard problems on the topic, and required a little bit of thinking rather than applying some known machinery.

iii) [ 5 ]

We transmit 000 for tails, and 111 for heads. For an error, at least two bits should be flipped. Probability of error if 0 is transmitted is given by

$$Pe_0 = 3(1-p_0)^2 p_0 + (1-p_0)^3$$

Probability of error if 1 is transmitted is given by

$$Pe_1 = 3(1-p_1)^2 p_1 + (1-p_1)^3$$

Note that $Pe_1 > Pe_0$. Probability of error is then given by $0.7Pe_0 + 0.3Pe_1$.